SCHOOL OF COMPUTER TECHNOLOGY

AASD 4008
Big Data Tools and Techniques

What is Apache Hadoop

The Apache™ Hadoop® project develops open-source software for reliable, scalable, distributed computing. [1]

The Apache Hadoop software library is a framework that allows for the distributed processing of large data sets across clusters of computers using simple programming models. It is designed to scale up from single servers to thousands of machines, each offering local computation and storage. ***In short!***

Hadoop is a framework that allows us to store and process large data sets in a parallel and distributed fashion.

Hadoop was designed to help Organizations manage terabytes

Apache Hadoop

[1] Apache Hadoop : https://hadoop.apache.org/

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Why Hadoop

Imagine this scenario

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- You have 5gb of data that needs to process. Data is stored and processed in a relational database, and you can query the data with ease.
- You had an exceptional year, more sales and more data, now you have 50 GB of data. Your computers / servers are still ok to deal with that.
- Now you gain another business lets say a new line of business ,the data is now 100 gigs, now you have reached the capacity of your current server, so you buy another one… now you gain another line of business that have chain of stores around the province , you know need to process large sum of survey data and thousands of transactions each second or perhaps you have to merge and analyze retail data with the weather data and the sensors data that is attached to the delivery truck the data volumes is now grown to 100 TB that includes the unstructured data coming from Facebook, twitter or other social media sites that have comments about the retailer you just acquired…. how do you process that large amount of data since?
- **Hadoop may be the answers , let's discuss**

Core Components / Base Modules

Hadoop Common	Contains libraries and utilities needs by other Hadoop modules
HDFS – Hadoop distributed File System	The distributed file system that stores and processes a vast amount of data that scales linearly. Self-healing, distributed file system for multi-structured data. Breaks files into blocks and store redundantly across the cluster.
Hadoop YARN	Resource manager! Application manager came new to hadoop 2.0

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1 Edureka : https://www.youtube.com/watch?v=1vbXmCrkT3Y&ab_channel=edureka%21 : https://www.edureka.co/

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How Hadoop Works

Traditional Scenario :

Let's assume a restaurant run by Mark

- He has one cook and one food shelf
- One cook is sufficient to prepare 4 orders per hour

In **traditional database** system, they generated structured data at a steady rate

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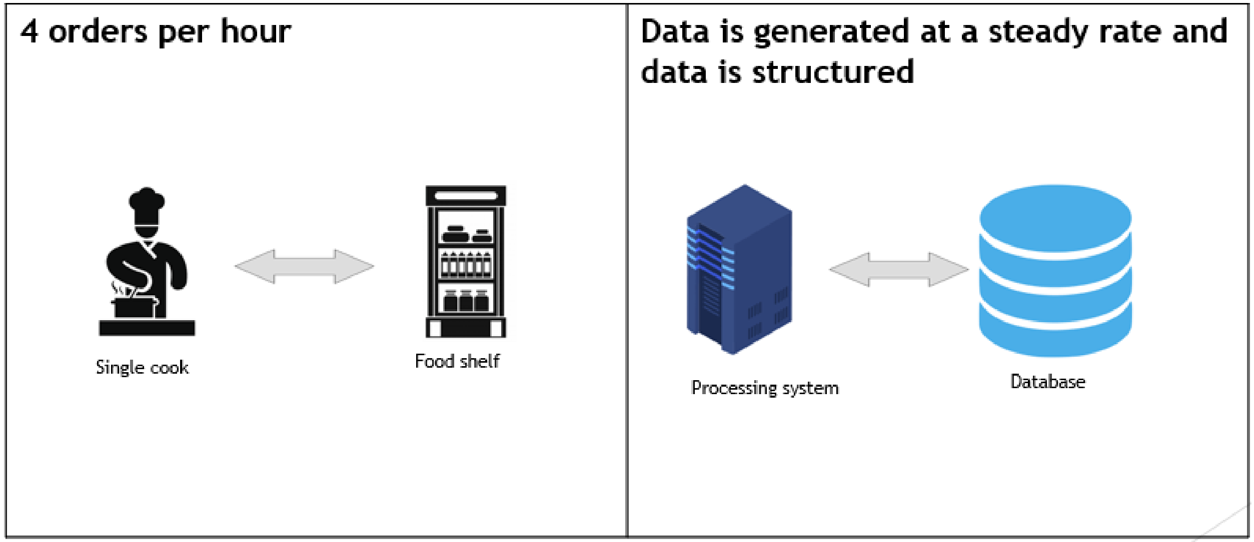

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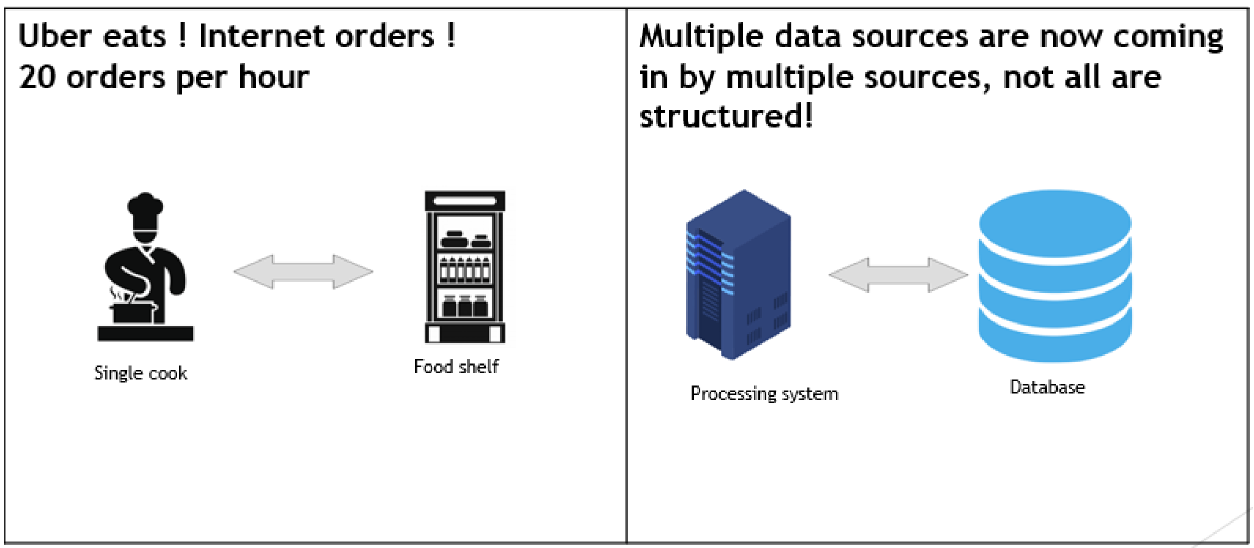
How Hadoop Works – Continued !

More Orders !

What do they do now ?

- 20 order are coming in through online orders

Multiple data sources are now coming in my multiple systems, not all sources are structured



[1]

[1] Edureka : https://www.youtube.com/watch?v=1vbXmCrkT3Y&ab_channel=edureka%21 : https://www.edureka.co/

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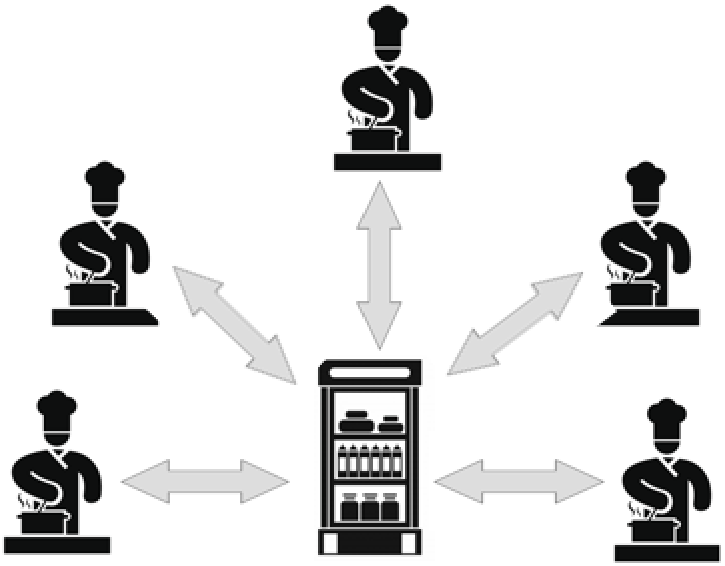
How Hadoop Works – Continued !

Solution ?

Hire multiple cook ?



No, that won’t help because

- Now the food shelf become bottleneck
- 2 cooks need the same item from the food shelf. One must wait for another.
- Another cook is waiting for the above cooks to finish before requesting an item from the food shelf

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1 Edureka : https://www.youtube.com/watch?v=1vbXmCrkT3Y&ab_channel=edureka%21 : https://www.edureka.co/

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How Hadoop Works – Continued !

Same goes for a traditional database, adding multiple servers won't help because.

- Data travelling from database to server will now become a bottleneck.
- Realtime processing will not be possible.

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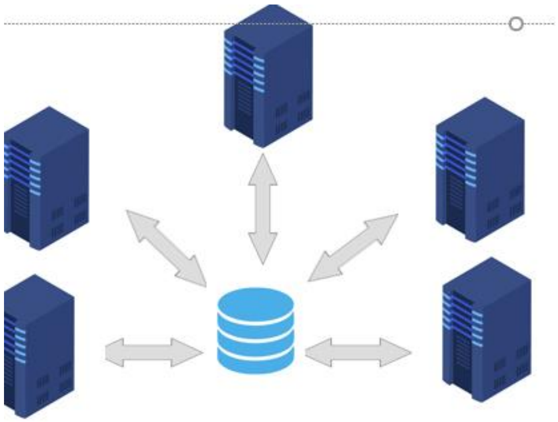

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How Hadoop Works – Continued !

Distributed and Parallel approach !

Let us take another approach to solve the issue, assume there are 20 orders of Fred Rice. 1

call it divide and conquer.

- Divided the order into steps.
- One cook will cook rice and other will cook meat and vegetable.
- Head cook with assemble.
- food shelve are distributed; each cook has his ingredients on his shelf which is near him/her.

benefits

- If one cook is sick, other can take his job.
- If one food shelf is not available, other are.
- Its scalable, on Christmas. Mark can hire more cooks and on off-peak time, fire them.

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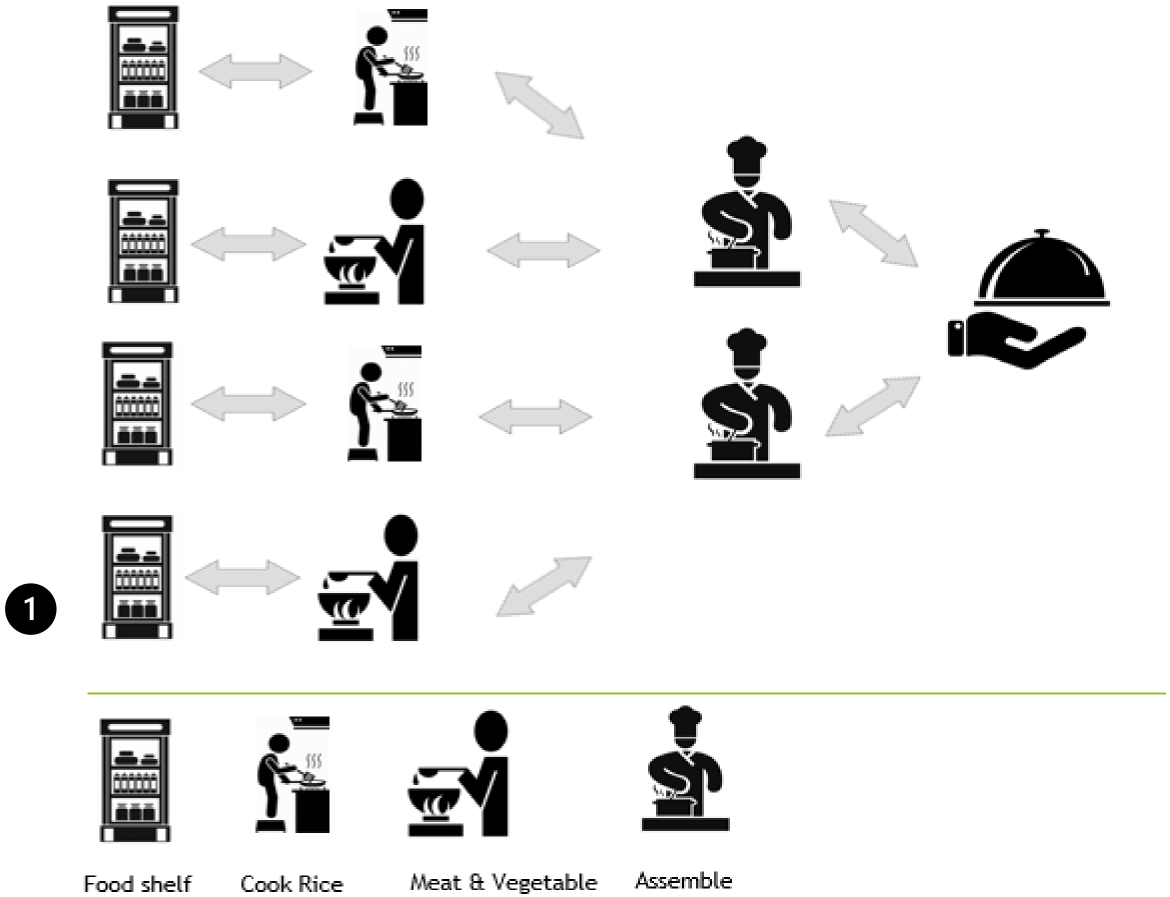
How Hadoop Works – Continued !

Distributed and Parallel approach !



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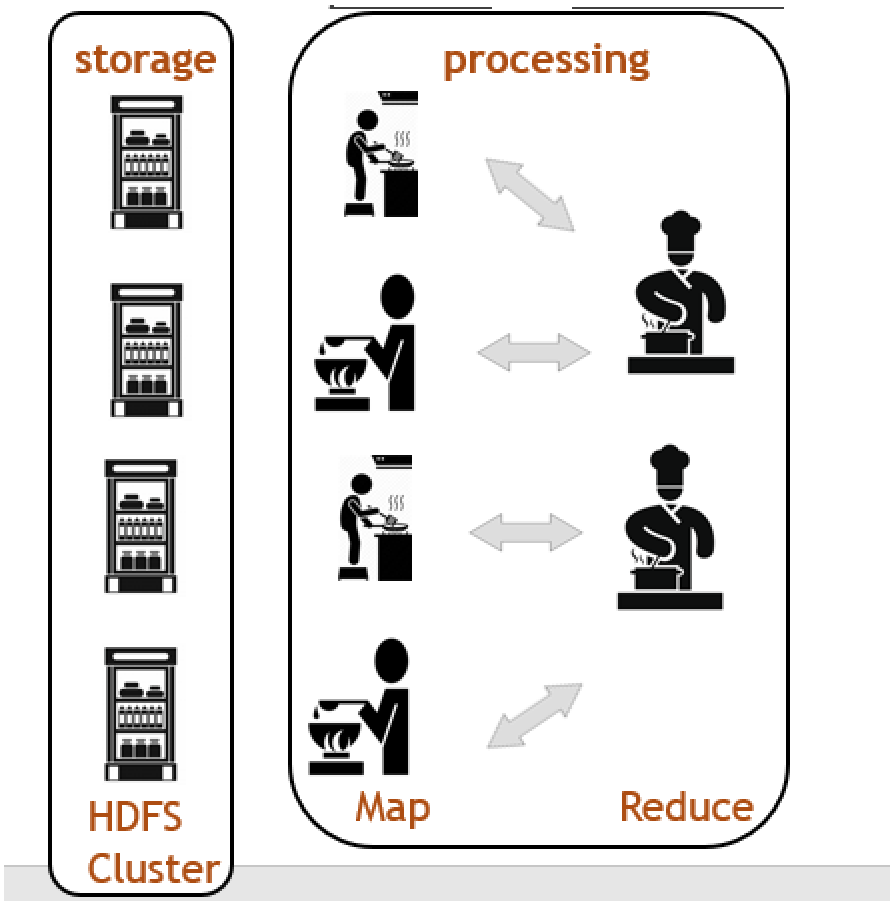
Relating the solution with Hadoop Framework

Hadoop is a framework that allows distributed[1] **storage** & **processing** of large datasets across clusters of computers. Its *stores* and process *large* datasets in *parallel* and *distributed* fashion.

We can relate the above example and solution to Apache Hadoop.

- The Food Shelf are **Storage** and Cooks are **Processing**.
- **Storage in Hadoop is HDFS Cluster**
- **Processing in Hadoop is Map → Reduce.**



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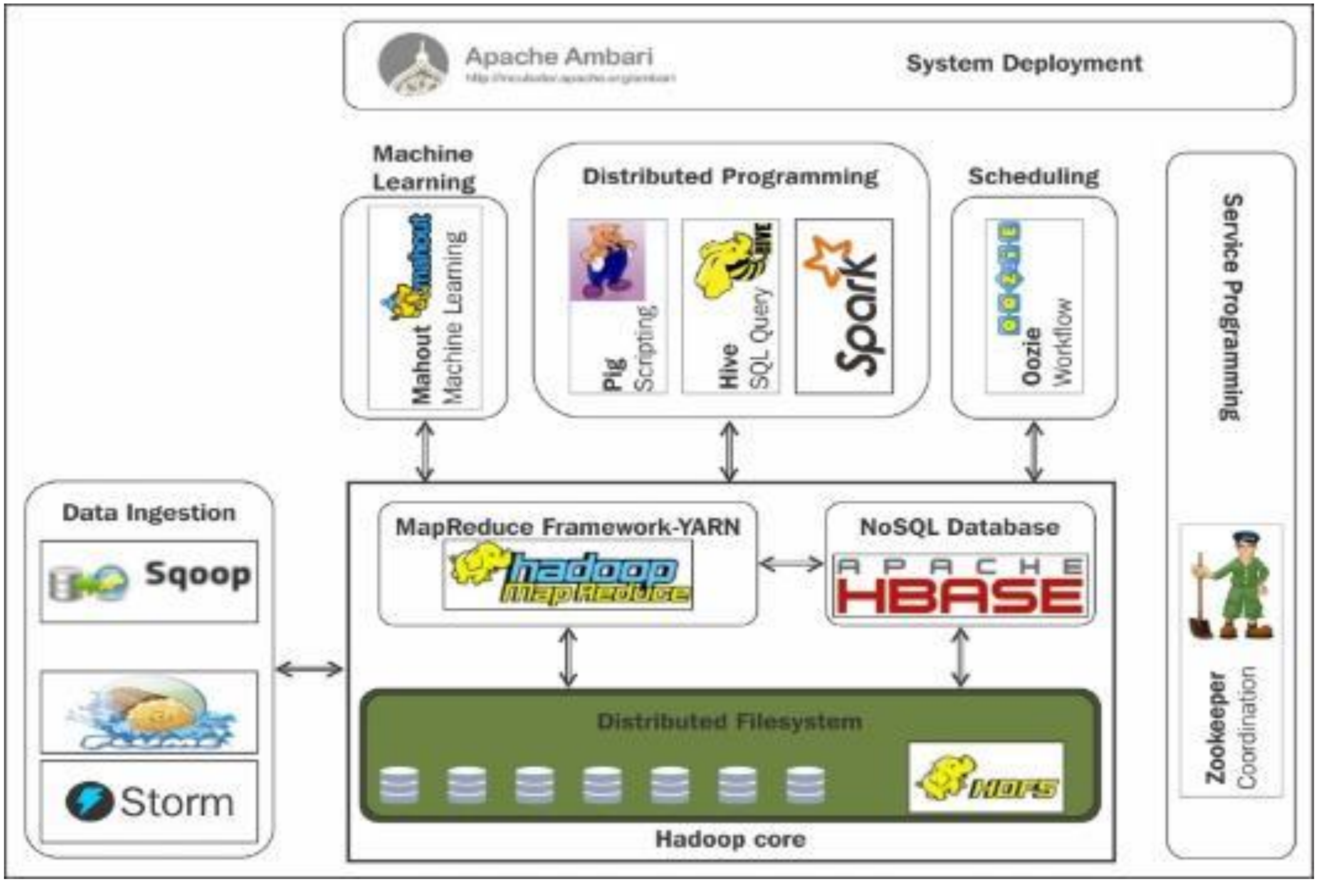
5 Pillars of Hadoop

[1]



1 Packtpub.com : https://subscription.packtpub.com/book/big-data-and-business-intelligence/9781784396688/2/ch02lvl1sec23/the-hadoop-ecosystem

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5 Pillars of Hadoop – Continued !

There are 5 pillars of Hadoop.

- Data Management
- Data Access / Service
- Data Governance/Integration
- Security
- Operations

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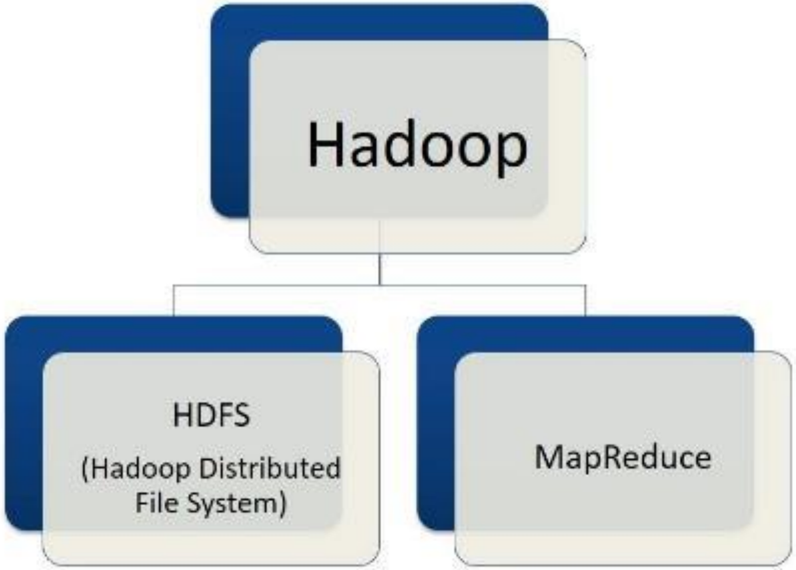
5 Pillars of Hadoop – Data Management

Data Management is the fundamental responsibility of Hadoop. We will discuss all of them in later sections.

HDFS	core technology for the efficient scale-out storage layer, and it runs across low-cost commodity hardware. HDFS stores data on commodity machines, providing very high aggregate bandwidth across the cluster. It's a JAVA-based file system that provides scalable and reliable data storage designed to span large clusters of commodity servers. HDFS solves big data storage problem by distributing data over different machines which are interconnected.
Map Reduce	MapReduce is the programing unit of Hadoop used parallels and distributed processing of the Big Data which is laying across Hadoop cluster. The fact that every machine in the Hadoop cluster processes the data it got is called MAP The process that combines the results is called reduce.
YARN	YARN is the pre-requisite for enterprise Hadoop. It provides resource management and pluggable architecture to enable various data access methods to operate on data stored in Hadoop with predictable performance and service levels. YARN is part of data management, it's not a data access tool, but its job is to make sure data is being accessed from HDFS through Data Access tools (hive, pig….)

5 Pillars of Hadoop – Data Management



Takes care of **storing and managing the data** in Hadoop Cluster

Process the data which is present within HDFS

5 Pillars of Hadoop – Data Access / Services - 1/3

Data access services is the fundamental responsibility of Hadoop. We will discuss all of them in later sections.

Apache Hive is the he most widely adopted data access technology, build on MapReduce framework, Hive is a data warehouse that enables easy data summarization and ad-hoc queries via SQL-Like interface for large datasets stored in HDFS.
Three 3 runtime choices Apache MapReduce, Apache TEZ and Apache Spark

Apache pig provides scripting capabilities. A platform for processing and analyzing large data sets. Pig consists of a high-level language (pig Latin) for expressing data analysis programs paired with MapReduce framework.Pig Latin allows you to write a data flow that describes how your data will be transformed (such as aggregate, join and sort).

MapReduce is a framework for writing applications that processes large amounts of structured and unstructured data in parallel across a cluster of thousands of machines, in a reliable and fault-tolerant manner.

5 Pillars of Hadoop – Data Access / Services - 2/3

Apache spark is ideal for in-memory data processing. It allows data scientists to implement fast, interactive algorithms for advanced analytics, such as clustering and classifications of datasets.

Apache Storm is a distributed real-time computation system for processing fast, large streams of data adding reliable real-time data processing capabilities to Hadoop.

Apache Tez generalizes the MapReduce paradigm to a more powerful framework for executing a complex DAG (Direct acyclic graph) of tasks for near real-time big data processing.

Apache Kafka is a fast and scalable publish - subscribe message system that is often used in place of traditional message brokers because of its higher throughput, replication, and fault tolerance.

Apache Slider is a framework for deployment of long-running data access applications in Hadoop. Slider leverages YARN's resource management capabilities to deploy those applications, to manage their lifecycles and scale them up or down

5 Pillars of Hadoop – Data Access / Services - 3/3

Apache Solr is the open-source platform for searches of data stored in Hadoop. Solr enables powerful full-text search and near real-time indexing on many of the world's largest internet sites.

Apache Mahout provides scalable machine learning algorithms for Hadoop, which aids with date science for clustering, classification and batch based collaborative filtering

Apache Accumulo is a high-performance data storage and retrieval system with cell-level access control. It is scalable implementation of Google's Big Table design that works on top of Apache Hadoop and Apache Zookeeper.

5 Pillars of Hadoop – Data Governance/Integration

Workflow manager provides workflows for data governance, allows you to easily create and schedule workflows and monitor workflow jobs. It is based on Apache Oozie workflow engine that allows users to connect and automate the execution of big data processing tasks into a defined workflow.

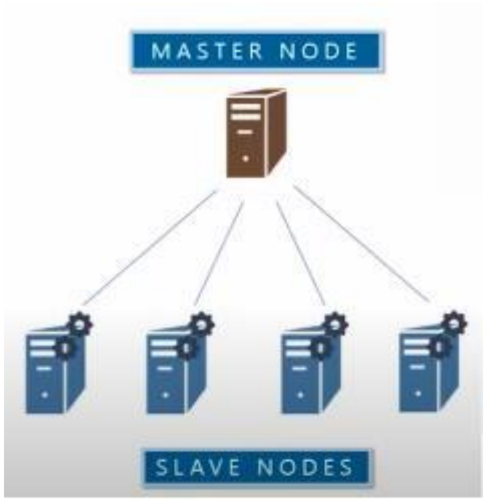
Apache Flume is an easy data ingestion program. Flume allows you to efficiently aggregate and move large amounts of data from many difference sources to Hadoop.

Apache Sqoop is an easy data ingestion and extraction. Sqoop is a tool that speeds and eases movement of data in and out of Hadoop. It provides a reliable parallel load for various, popular enterprise data sources

Hadoop Core Components – Continued – HDFS

HDFS has 3 nodes

- Name Node
  - Also called **Master Node**
  - Maintains and managed data nodes (Slave Nodes)
  - Records metadata
    - Information about data blocks
    - Locations of blocks stored
    - Size of files/permissions/hierarchy
- Data Node
  - Also called **Slave Nodes**
  - **Slave demons**
  - Stores actual data
  - Serve read and write requirements from the client
- Secondary Node
  - Responsible for checkpointing



Hadoop Core Components – Continued – HDFS

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Checkpointing

Secondary NameNode plays a crucial role in enhancing the overall fault tolerance and reliability HDFS. While it is often referred to as the Secondary NameNode, it is not a backup or standby for the primary NameNode. Instead, it assists the primary NameNode in managing the HDFS by performing periodic checkpoints.

Checkpointing is the process of combining the fsimage (file system image) and the edits log files generated by the primary NameNode.

- **The fsimage represents the state of the file system at a particular point in time**, while the **edits log contains the incremental changes made to the file system since the last checkpoint.**

1 Data-Flair : https://data-flair.training/blogs/spark-streaming-checkpoint/

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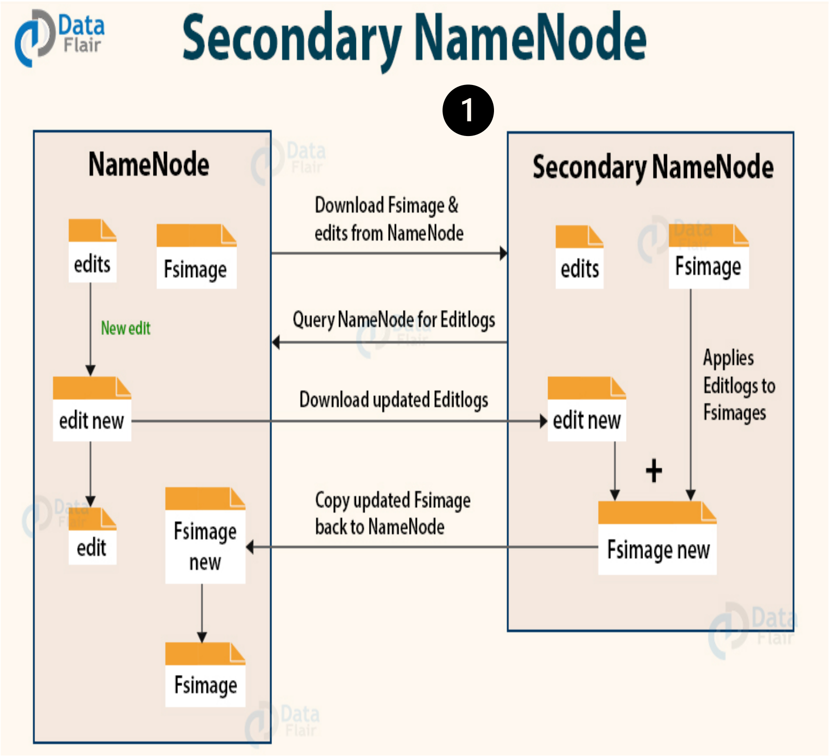
Hadoop Core Components – Continued – HDFS

Checkpointing

- Check pointing is combining edit logs with FSImage
- Edit log is in REM, it has to be smaller for quick response.
- Secondary Name Node takes over the responsibility of check pointing, therefore making Name Node more available.
- Allows Faster Fail over s it prevents edit logs from getting too huge.
- Check pointing happens periodically (default 1 hour)
- To set up a new name node (if name node fails), Hadoop can copy the fsimage form secondary name node



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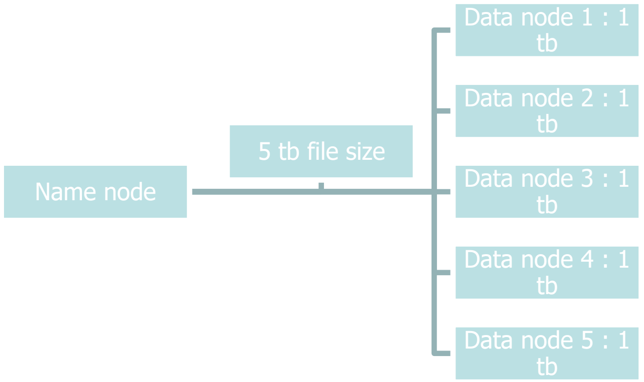
Hadoop Core Components – Continued – HDFS

Data Block : Each file is stored on HDFS as blocks of 128mb (default size) . Following are some of the benefits

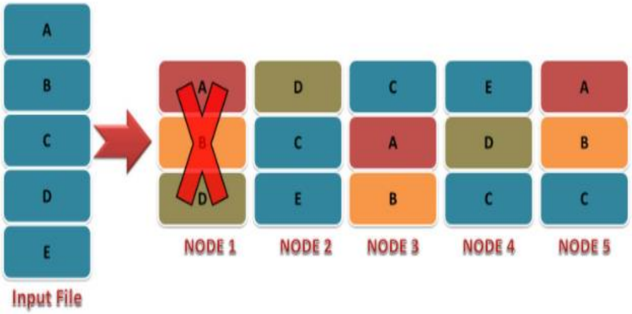
- **Big Data Storage :** Let us assume we have a Hadoop cluster with 6 machines, one is a name node, and 5 are data nodes. The capacity of data nodes is 1 Tb each. If we were to store a file with 5 TB, Hadoop would distribute that into 5 nodes. Hadoop provides us an abstraction of one high-end machine with a capacity of 5 TB. while it might take 5 minutes to store 5TB of data in a high-end device, but in Hadoop, it will take 1 minute.



Hadoop Core Components – Continued – HDFS

Processing large datasets : Let us assume we have a high-end computer with a processing speed of 5 minutes for 5 TB data. With HDFS, that data will be processed and stored in a parallel and distributed fashion, so if we have five nodes, it will take 1 minute each. The data processing is now done in 1 minute as opposed to 5.

Fault Tolerance : What if one node crashes, now a part of your file is not available? Remember, we are dealing with commodity hardware in Hadoop, and a node is likely to crash, and Hadoop is ready for that!
Hadoop uses a Replication factor. all data block has 3 copies on 3 different nodes across the cluster. So, every data block will be replicated in the Hadoop cluster 3 times, so even if one node is off, the data will be available on 2 other nodes, and in the worst-case scenario, if 2 nodes go down, 1 node is still available. Once we add more healthy nodes, Hadoop will start replicating again and this step is called auto-healing!



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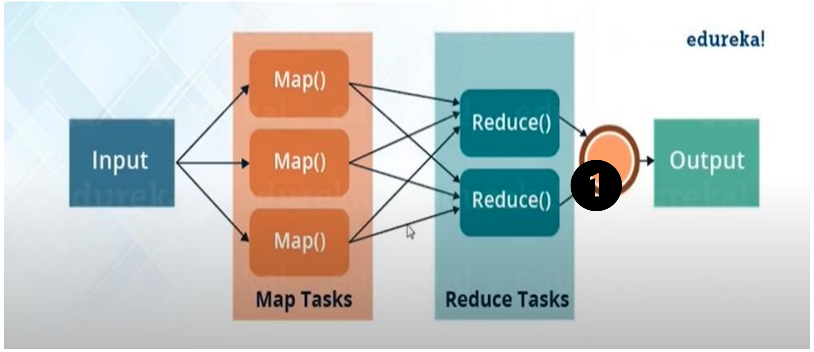
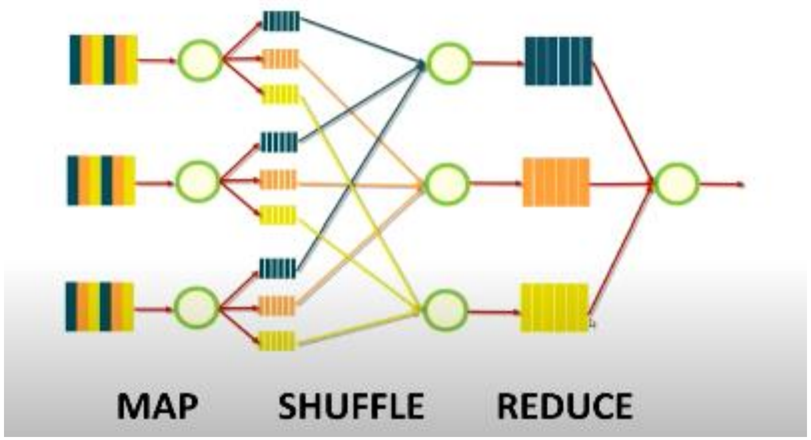
Hadoop Core Components – Continued – MapReduce

MapReduce

- MapReduce is a distributed and parallel framework to process large datasets.
  - Two distinct tasks, **Map** and **Reduce**
  - **Map**: Read and process data into key-value pairs as an intermediate output,
  - then pass on those results to the **reducer.**





1 Edureka.co : https://www.edureka.co/

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Hadoop Core Components – Continued – MapReduce

How MapReduce Work

Let us take a **word count program**

- There are two input files, each with two lines, as shown in the diagram below.
- Each line will be passed to an **individual mapper** instance.
- **Map Key-value splitting** step will collect each key (apple, oranges...) and assign them 1 as a value; hence, they now become key-value pairs.
- The next step **sort and shuffle** will collect Key-Value pairs from the splitting step and sort them by keys.
- the last step for the **reducer** to add all values for the same key and pass them on to the client.

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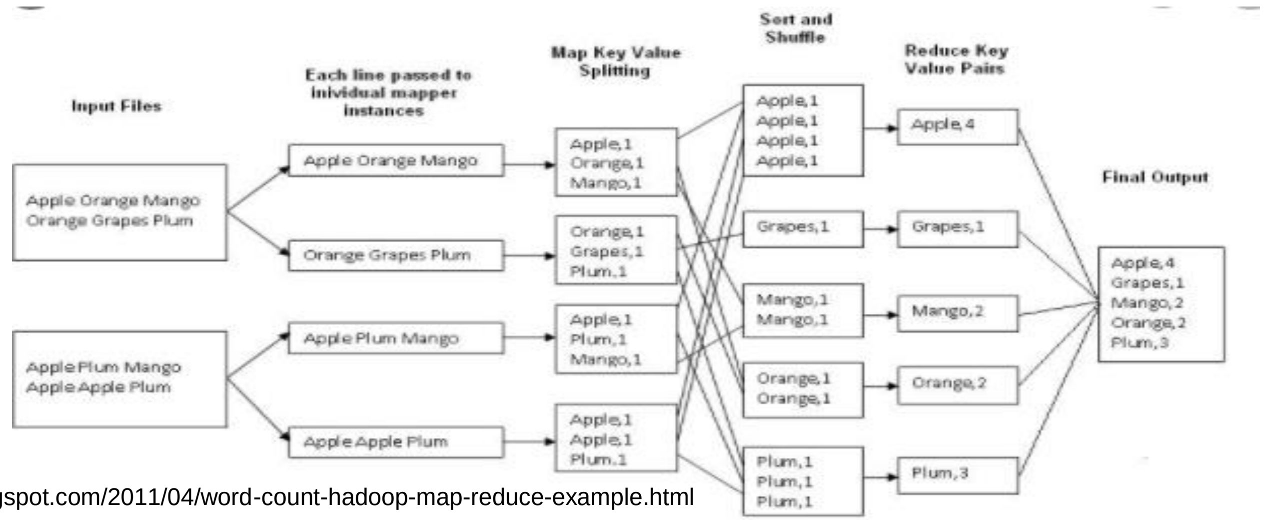
- Assignment : https://www.guru99.com/create-your-first-hadoop-program.html



1 Kick Start Hadoop : http://kickstarthadoop.blogspot.com/2011/04/word-count-hadoop-map-reduce-example.html

Please also refer to references slide at the end knowledge or reference

Hadoop Core Components – Continued – YARN

YARN

YARN manages computing resources in clusters and uses them to schedule user's applications. YARN is a next-generation framework for Hadoop data processing extending MapReduce capabilities by supporting non-MapReduce workloads associated with other programming models (TEZ and Spark)

The most significant limitation with MapReduce programming is that map and reduce jobs are not stateless. This means that reduce jobs must wait for map job to complete first. This limits maximum parallelism, and therefore YARN was born as a generic resource management and distributed application framework!

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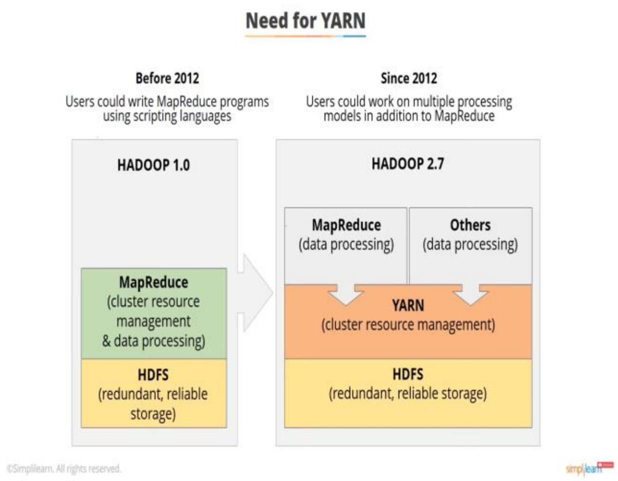
- Monitors and manages workloads.
- establish a multi-tenant environment.
- Takes over the job of cluster resource management from MapReduce.
- Manages the high availability features of Hadoop.
- Implement security controls.



https://www.youtube.com/watch?v=KqaPMCMHH4g&ab_channel=Simplilearn
https://www.youtube.com/watch?v=fCoivEBDmYA&ab_channel=Simplilearn

1 Simplilearn.com : https://www.simplilearn.com/

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Hadoop Core Components – Continued – YARN - continued

Hadoop YARN is crucial for efficient resource management and job scheduling in a Hadoop cluster. It enables scalability, flexibility, and fault tolerance, allowing various types **of applications to coexist and share resources optimally**. With dynamic resource allocation and fault tolerance mechanisms, YARN maximizes cluster utilization, improves reliability, and accommodates diverse workloads. Its extensibility further expands the capabilities of the Hadoop ecosystem, making YARN a vital component for processing big data effectively. 1

1 Simplilearn.com : https://www.simplilearn.com/

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Credits and Citations

- Some material and image are taken form Edureka website and their YouTube tutorials.

  https://www.edureka.co/

- Some material and image are taken form simpliearn website and their YouTube tutorials.

  https://www.simplilearn.com/

- https://www.cselectricalandelectronics.com/big-data-evolution-of-big-data-benefits-of-big-data-opportunities/

- zdnet.com (https://www.zdnet.com/article/ten-examples-of-iot-and-big-data-working-well-together)
- Big Data Framework.org (https://www.bigdataframework.org/business-drivers-for-big-data)
- Apache Pig https://pig.apache.org/